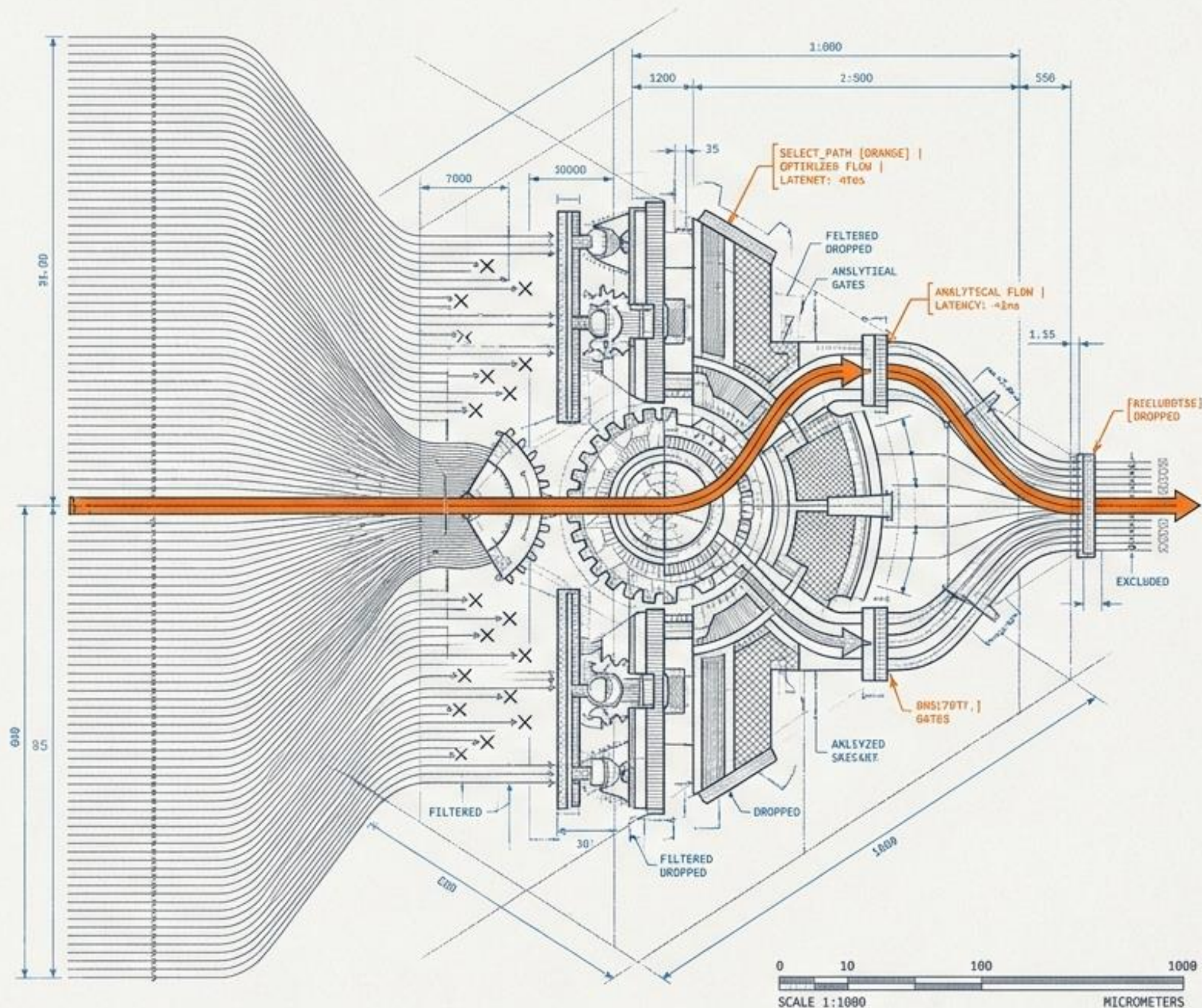


NeuralToolRouter

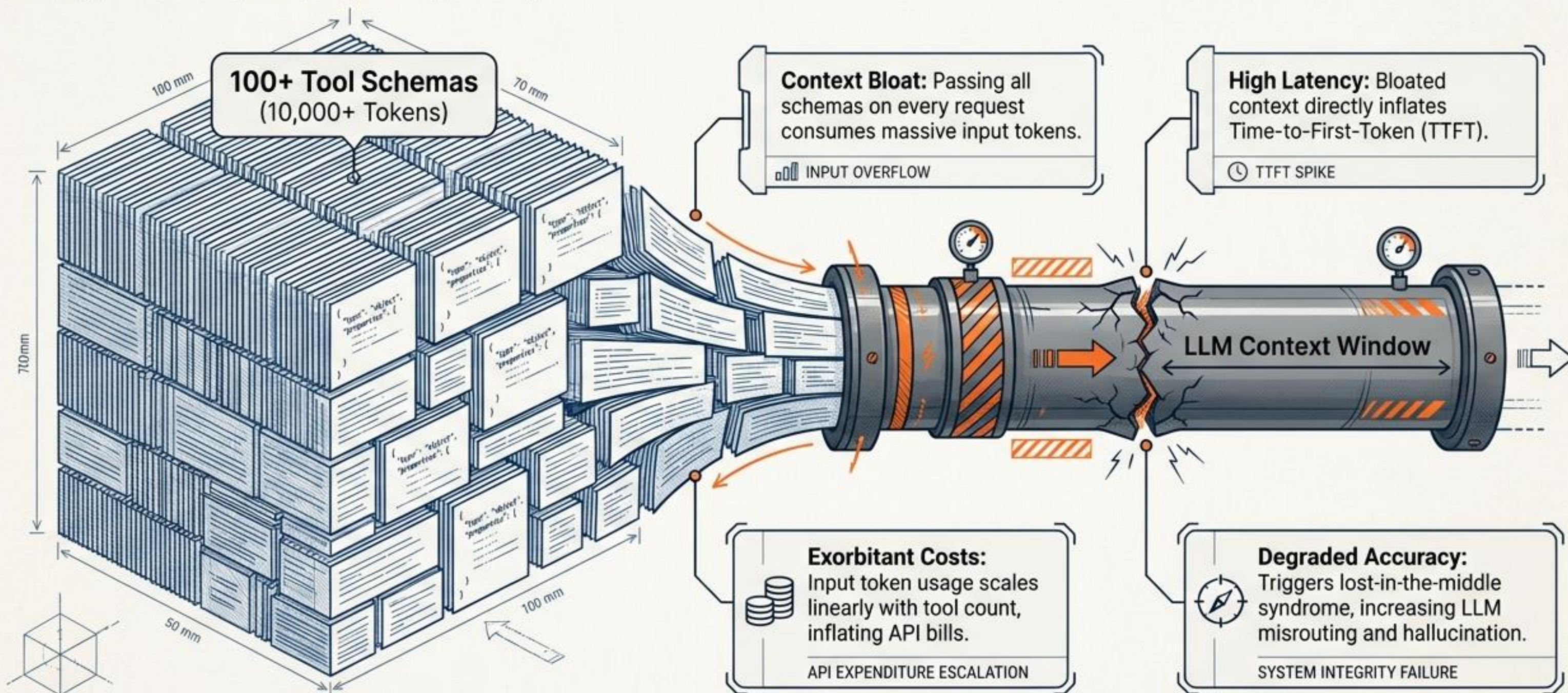
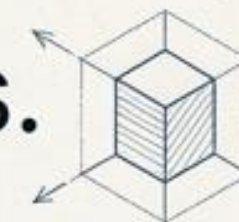
Scaling Agentic AI
architectures via
decoupled tool retrieval.

A blueprint for **eliminating context
bloat** and **reducing API latency**.



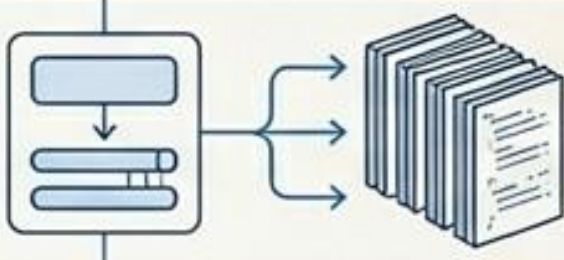
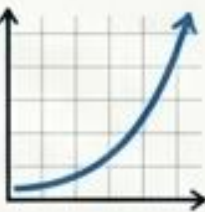
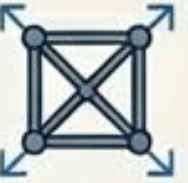
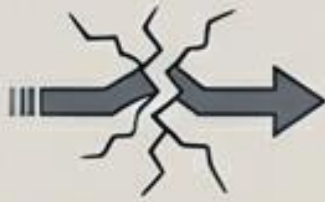
The Agentic Scaling Wall breaks standard architectures.

A High-Fidelity Engineering Analysis



Decoupling retrieval from execution solves context bloat.



Dimension	The Legacy Approach. Harsh restrictive border, constrained feel.	NeuralToolRouter (RAG-for-Tools). Generous padding, expansive framing, crisp blue typography.
Architecture	Inject All Schemas 	Decoupled Semantic Retrieval 
Context Load	10,000+ Tokens 	500–1,000 Tokens (Top-K only) 
Scalability	⚠ Breaks at ~50 tools 	 O(1) constant-time scaling to 1,000+ tools 
Fallback 	⊗ High failure rate 	Dynamic search_available_tools self-correction if initial retrieval misses  SELF-CORRECTING

Targeted retrieval yields massive performance dividends.



A bar chart with approximately 20 bars of varying heights, showing a significant decrease in context length from left to right. The first few bars are tall, while the last few are very short, illustrating a 90%+ reduction.

90%+ Context Reduction

Payload condensed strictly to **essential tools**.



A diagram consisting of a series of horizontal and vertical lines forming a descending staircase shape, representing a step-by-step reduction in latency from 4.5s down to 1.5s.

1.5s – 4.5s Faster TTFT

Net latency drops dramatically, easily absorbing the **micro-overhead** of vector retrieval.



A line graph showing a curve that starts at a high point on the left and decays exponentially towards the right, representing a ~90% cost saving.

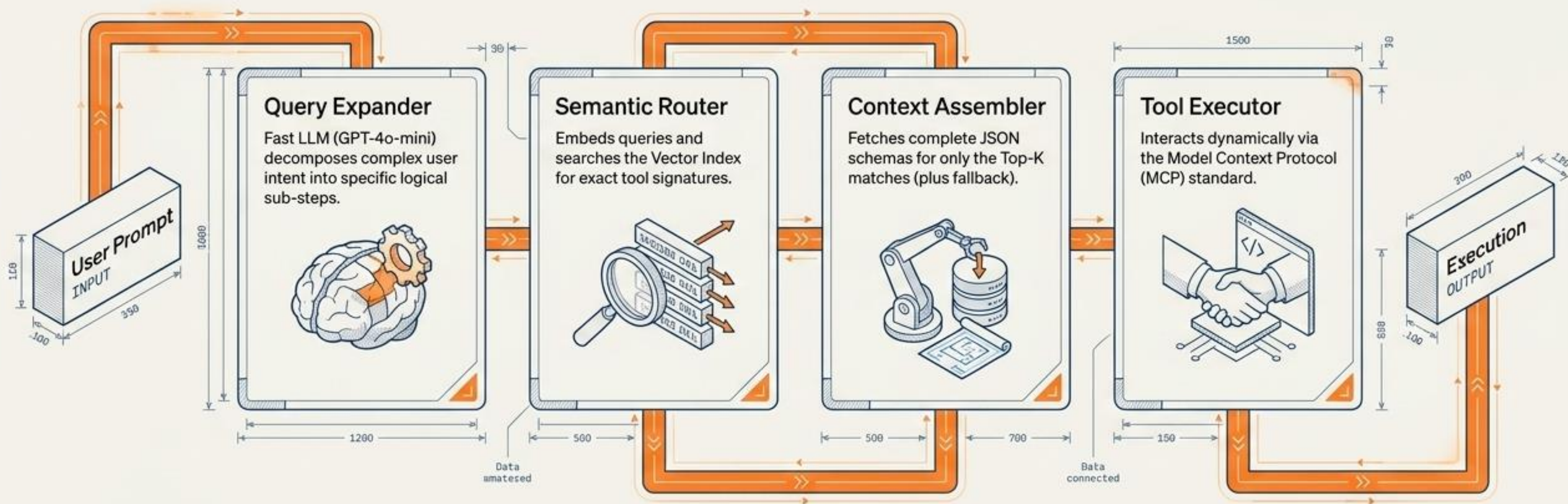
~90% Cost Savings

Direct, proportional reduction in LLM input token billing.

$O(1)$ Search Latency

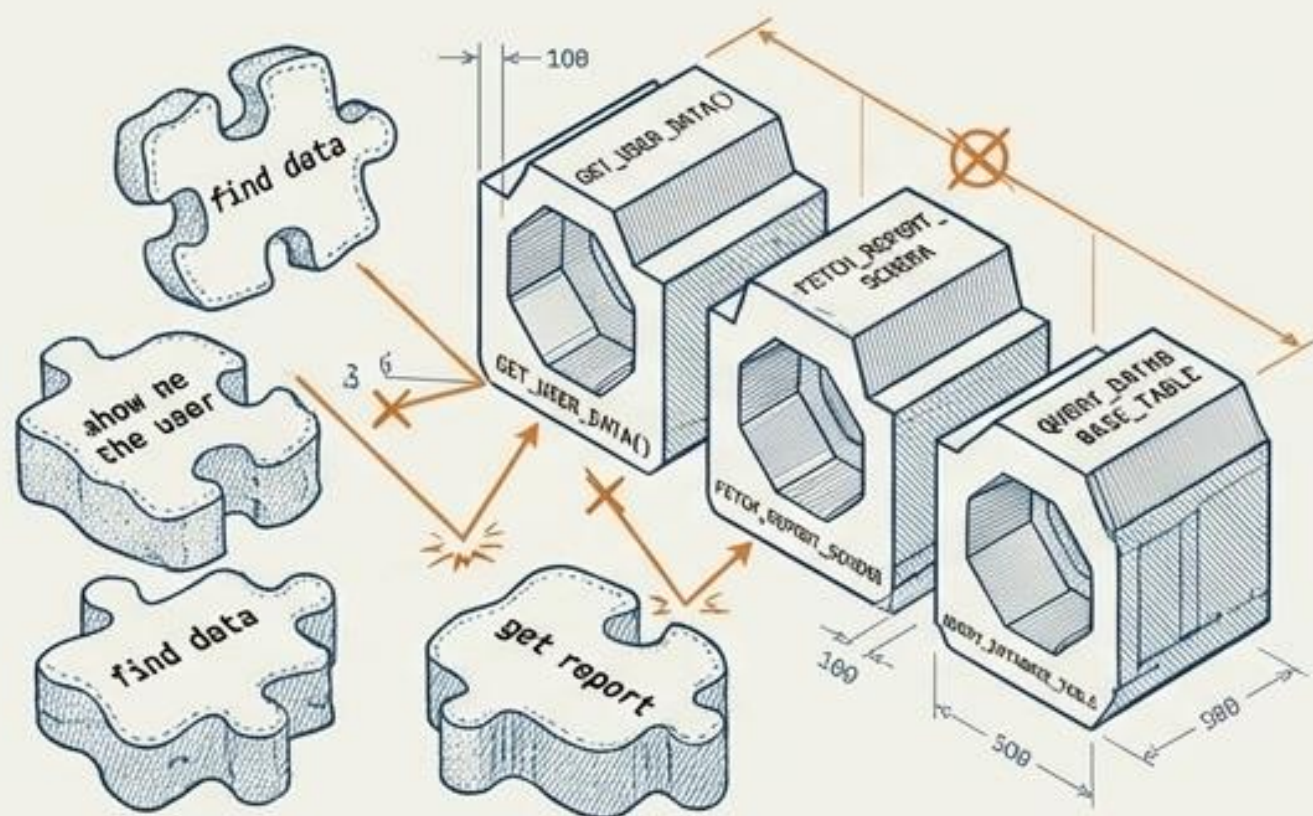
Instantaneous lookups powered by high-speed vector retrieval (FAISS/ChromaDB).

Anatomy of the NeuralToolRouter execution flow.



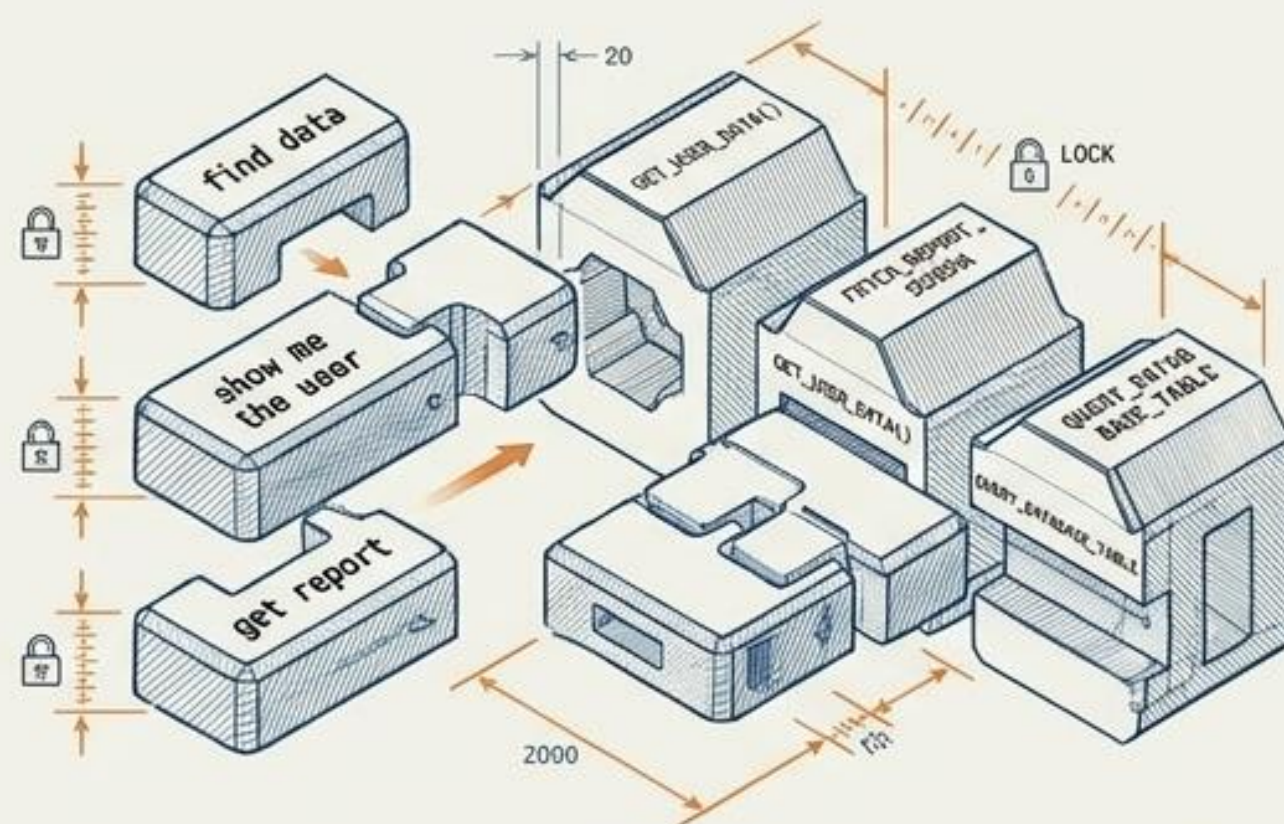
Bridging the semantic gap between natural language and code.

The Zero-Shot Problem



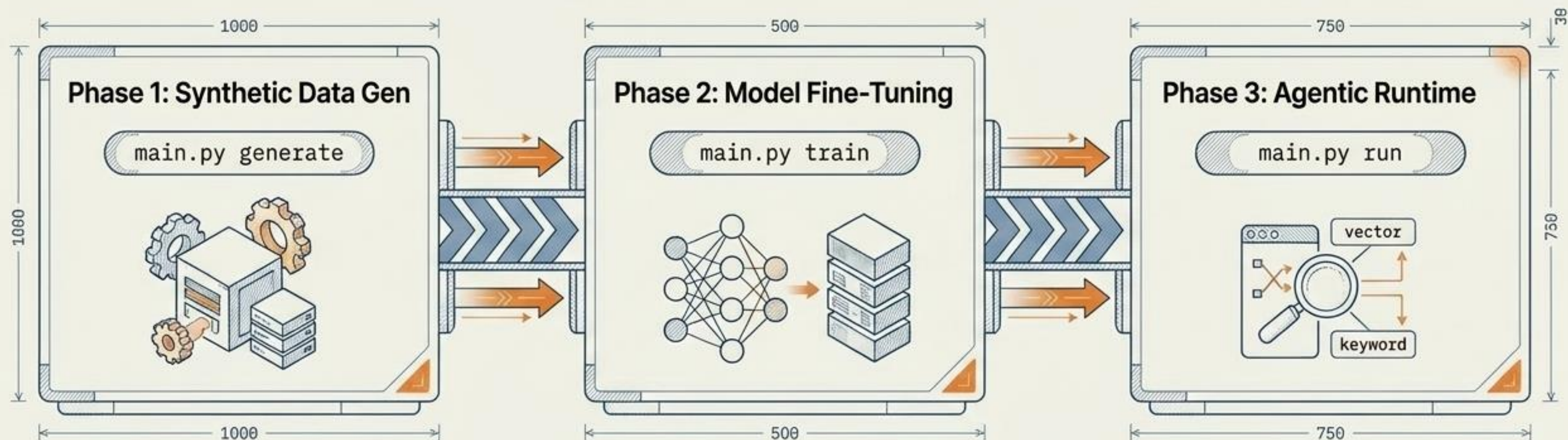
Out-of-the-box embedding models fail to accurately map abstract human requests to strict programming terminology.

The Fine-Tuned Solution



Training via contrastive learning on your unique tool definitions ensures enterprise-specific terminology maps precisely, guaranteeing production-grade accuracy.

A production-ready pipeline for domain-specific accuracy.

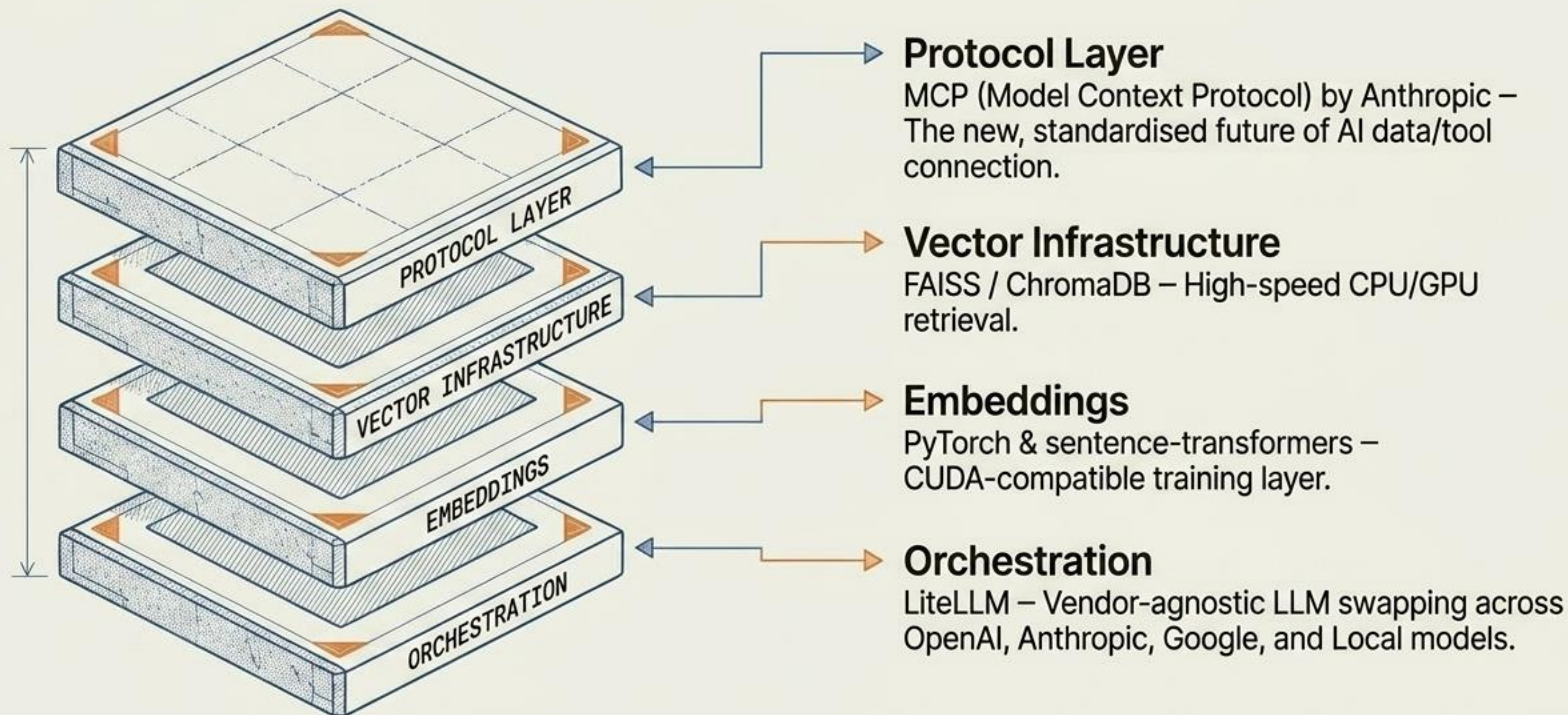


Phase 1: Synthetic Data Gen:
Connects to MCP servers to fetch schemas; uses a Teacher LLM to generate diverse, realistic queries.

Phase 2: Model Fine-Tuning:
Applies contrastive learning on synthetic data to train a lightweight model (e.g., all-MiniLM-L6-v2); builds the FAISS/Chroma index.

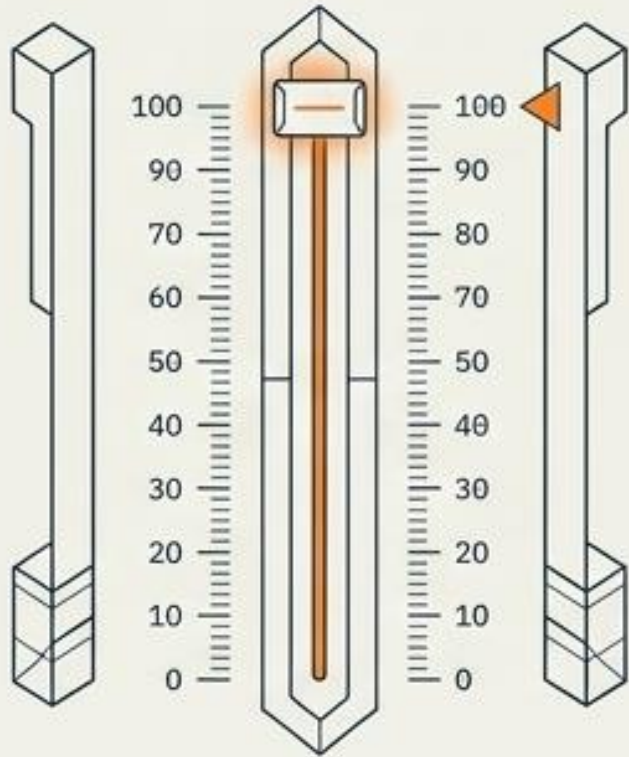
Phase 3: Agentic Runtime:
Live execution utilising Hybrid Retrieval (Semantic Vector + BM25) for precision routing.

Built entirely on bleeding-edge open standards.

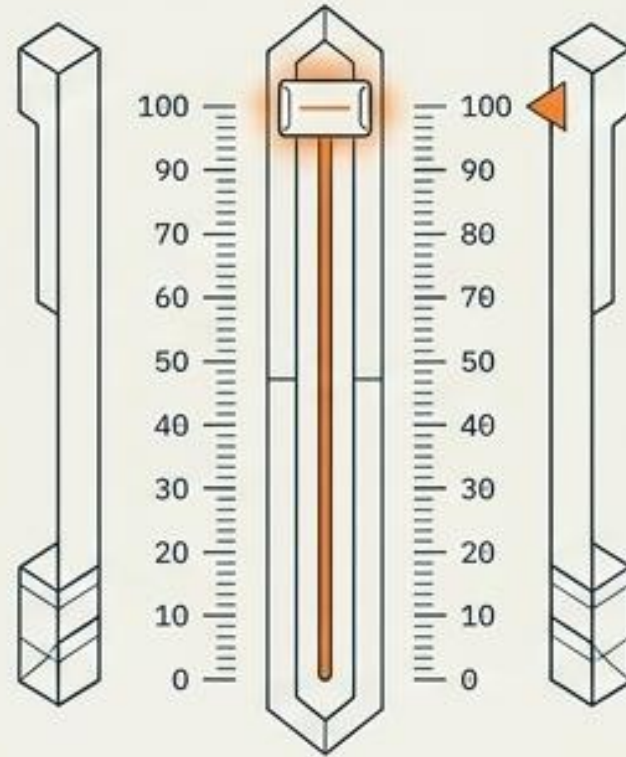


Architectural dials to balance speed, accuracy, and cost

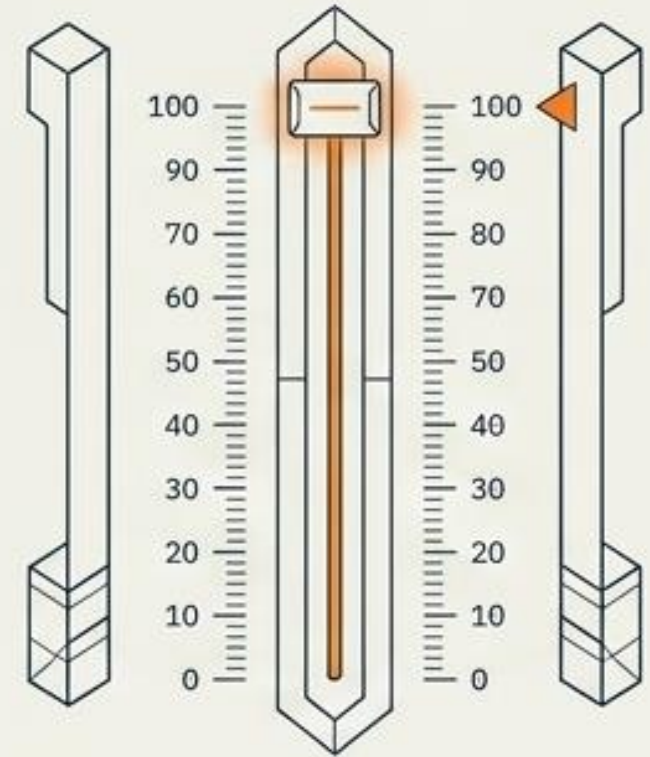
SPEED



ACCURACY



COST



Speed Optimisation

- Switch to paraphrase-MiniLM-L3-v2
- Reduce Top-K to 2
- Disable query expansion
- Deploy on CPU-based FAISS

Accuracy Optimisation

- Upgrade to all-mpnet-base-v2
- Increase Top-K to 5
- Maximise training epochs
- Enforce Hybrid Retrieval

Cost Optimisation

- Swap Expansion LLM from cloud providers to local open-source models or GPT-4o-mini

Key Insight

Bypass standard runtime limitations without rebuilding your agents. Simply replace standard tool arrays or your existing `ToolNode` with a `SemanticRouter.retrieve_tools()` call. Pass only the resulting schemas into your Agent's prompt context.



Future-proof your agentic infrastructure today.

```
> git clone https://github.com/sinny777/neural-tool-router
```



Review [ARCHITECTURE.md](#) to begin the 3-phase deployment setup.

Coming Next: Active learning from user feedback, multi-modal routing, and hierarchical tool logic.